Day 110 | Unit 08 | Teacher plan

Solve Area, Surface Area, and Volume Problems

Justify Polygon Area with Coordinates

Standards: 6.G.A.3, 6.G.A.1

Learning target: I can justify Polygon Area with Coordinates, and explain and check my reasoning.

enVision Topic 7: Solve Area, Surface Area, and Volume Problems | Spectrum companion Chapter 6

0–5 minutes / Launch

Read the target. Ask for an estimate or a sketch before calculating. Connect the new idea to the previous lesson.

5–17 / Model

Use Worked Example 1, then Example 2. Ask students to name the next step and explain a choice. Plot vertices in order and connect them, including the last to the first. Equal x-values make a vertical side; equal y-values make a horizontal side. Count or subtract coordinates to get the side lengths. Sketch the figure and label every given length. Split the region into nonoverlapping rectangles and triangles, or subtract a cutout from a larger rectangle. Add only the areas that belong to the figure.

17–42 / Practice

Assign questions 1–10 in order. Use 1–3 for foundations, 4–7 for application, and 8–10 for state-format response and reasoning. The example button provides a separate question with a worked method.

42–50 / Discuss and check

Compare representations. Review a selected answer and a plausible error. Use question 10 as an exit discussion; ask students to check units, scale, or substitution.

Support and extension

Offer number lines, fraction strips, grid paper, or a table template as appropriate. Read multi-part directions one part at a time. For extension, ask for another representation or a counterexample.

Companion and state alignment

Use the matching skill in the edition you own. No unverified publisher page numbers are assigned. AASA references below identify related public sample items by standard and format; all course questions are original.

AASA sample item references: 23

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01 / Learn

Plot vertices in order and connect them, including the last to the first. Equal x-values make a vertical side; equal y-values make a horizontal side. Count or subtract coordinates to get the side lengths. Sketch the figure and label every given length. Split the region into nonoverlapping rectangles and triangles, or subtract a cutout from a larger rectangle. Add only the areas that belong to the figure.

Worked example 1

Plot and connect (0,0), (5,0), (5,9), (0,9). Find the side lengths and area.

1. Plot vertices in order and connect them, including the last to the first.

2. Equal x-values make a vertical side; equal y-values make a horizontal side.

3. Horizontal length 5 - (0) = 5

4. vertical length 9 - (0) = 9

5. area = 5 × 9.

6. State the result in the form requested: Sides 5 and 9; area 45 square units.

Answer: Sides 5 and 9; area 45 square units

Worked example 2

A 7 m by 9 m rectangle has a right-triangle corner removed with perpendicular legs 4 m and 5 m. Part A: enter the removed area. Part B: enter the remaining area and justify subtraction.

1. Sketch the figure and label every given length.

2. Outer area = 63.

3. Triangle = 4 × 5/2 = 10.

4. Subtract the removed region exactly once.

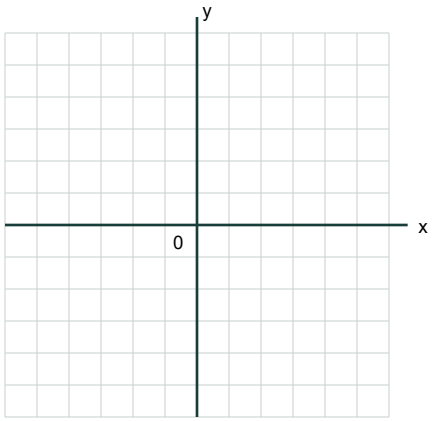
Answer: Removed 10 m²; remaining 53 m²

Day 110 | Your turn

Work in order: foundation → application → explanation. Show calculations and label units.

01 / Constructed response

Plot and connect (0,0), (11,0), (11,5), (0,5). Find the side lengths and area.



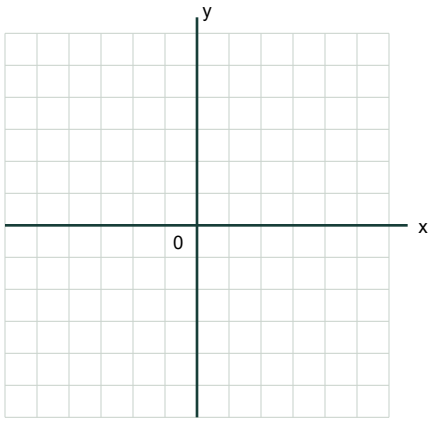
Label the scale and units.

02 / Constructed response

An L-shaped garden is a 13 m by 10 m rectangle with an 8 m by 5 m corner rectangle removed. Sketch and find the garden area.

03 / Constructed response

Plot and connect (0,0), (11,0), (11,10), (0,10). Find the side lengths and area.



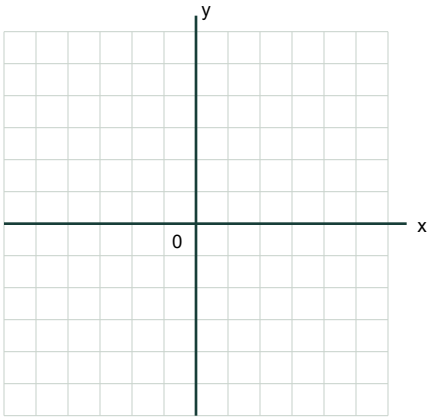
Label the scale and units.

04 / Constructed response

An L-shaped garden is a 6 m by 10 m rectangle with an 3 m by 7 m corner rectangle removed. Sketch and find the garden area.

05 / Constructed response

Plot and connect (-8,-4), (3,-4), (3,3), (-8,3). Find the side lengths and area.



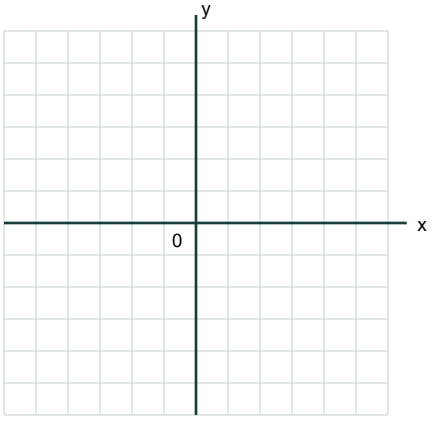
Label the scale and units.

06 / Constructed response

An L-shaped garden is a 6 m by 5 m rectangle with an 3 m by 2 m corner rectangle removed. Sketch and find the garden area.

07 / Constructed response

Plot and connect (-2,-4), (6,-4), (6,6), (-2,6). Find the side lengths and area. Find the perimeter too.



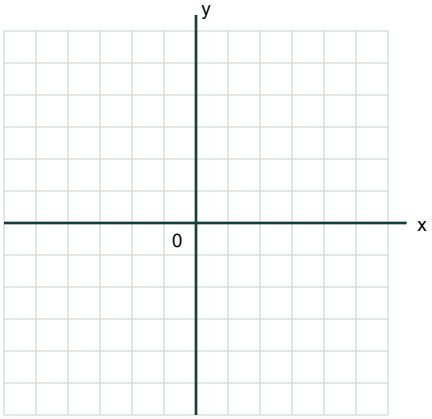
Label the scale and units.

08 / Two-part numeric entry

A 9 m by 9 m rectangle has a right-triangle corner removed with perpendicular legs 3 m and 6 m. Part A: enter the removed area. Part B: enter the remaining area and justify subtraction.

09 / Coordinate plotting and two-part response

A rectangle has vertices (-2, -2), (4, -2), (4, 2), and (-2, 2). Plot and connect them. Part A: find its side lengths. Part B: enter its area in square units.



Label the scale and units.

10 / Constructed response

A 10 m by 11 m rectangle has a right-triangle corner removed with perpendicular legs 4 m and 7 m. Sketch the remaining polygon and find its area.

Day 110 | Unit 08 | Answer key

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1. Sides 11 and 5; area 55 square units

Horizontal length 11 - (0) = 11; vertical length 5 - (0) = 5; area = 11 × 5.

6.G.A.3, 6.G.A.1 | Constructed response

2. 90 m²

Outer area 130 minus cutout 40 = 90.

6.G.A.1 | Constructed response

3. Sides 11 and 10; area 110 square units

Horizontal length 11 - (0) = 11; vertical length 10 - (0) = 10; area = 11 × 10.

6.G.A.3, 6.G.A.1 | Constructed response

4. 39 m²

Outer area 60 minus cutout 21 = 39.

6.G.A.1 | Constructed response

5. Sides 11 and 7; area 77 square units

Horizontal length 3 - (-8) = 11; vertical length 3 - (-4) = 7; area = 11 × 7.

6.G.A.3, 6.G.A.1 | Constructed response

6. 24 m²

Outer area 30 minus cutout 6 = 24.

6.G.A.1 | Constructed response

7. Sides 8 and 10; area 80 square units; perimeter 36 units

Horizontal length 6 - (-2) = 8; vertical length 6 - (-4) = 10; area = 8 × 10.

6.G.A.3, 6.G.A.1 | Constructed response

8. Removed 9 m²; remaining 72 m²

Outer area = 81. Triangle = 3 × 6/2 = 9. Subtract the removed region exactly once.

6.G.A.1 | Two-part numeric entry

9. 6 by 4 units; area 24 square units

Horizontal length = 2 + 4; vertical length = 2 + 2. Multiply 6 × 4 = 24.

6.G.A.3, 6.G.A.1 | Coordinate plotting and two-part response

10. 96 m²

Whole rectangle 110; remove triangle 14; difference 96.

6.G.A.1 | Constructed response